

TASK 11

1. Generative AI (GenAI)

Ye poori AI ki ek barri branch hai. Iska kaam sirf purana data dikhana nahi, balkay **naya content create karna** hai (chahe wo text ho, image ho ya music).

Jese ChatGPT aur Midjourney .

2. GANs (Generative Adversarial Networks)

Ye GenAI ki ek purani technique hai. Ismein do AI models aapas mein muqabla karte hain. Ek nakli cheez banata hai (Generator) aur doosra pakarta hai ke ye real hai ya fake (Discriminator). Is "chor-sipahi" ke chakkar mein AI itna perfect ho jata hai ke wo real jaisi images banane lagta hai.

3. LLMs (Large Language Models)

Jaise GPT-4 ya Llama. Inhone poori internet ki books aur articles parh rakhe hote hain, isliye ye insano ki tarah baat kar sakte hain aur sawalon ke jawab de sakte hain.

4. Vector

AI ko English samajh nahi aati, usay sirf numbers samajh aate hain. Jab hum kisi lafz ya sentence ko **numbers ki ek list** mein badal dete hain , toh us list ko hum "Vector" kehte hain.

5. VectorDB (Vector Database)

Ye ek special kism ka store-room hai. Aam database (SQL) mein hum naam ya roll number dhoondte hain, lekin VectorDB mein hum **milte julte matlab** dhoondte hain. Ye vectors ko store aur search karne ke liye use hota hai.

6. FAISS (Facebook AI Similarity Search)

Ye Facebook ki banayi hui ek super-fast library hai. Jab hamare paas lakho vectors hon, toh FAISS un mein se sabse fast tareeke se match dhoond kar lata hai. Ye samajh lein ke ye VectorDB ka "engine" hai.

7. RAG (Retrieval-Augmented Generation)

Retrival k liye yani memory rakhne k liye use hota h. LLMs kabhi kabhi purani baat bhool jate hain ya ghalat jawab dete hain. RAG ek tareeka hai jis mein hum LLM ko pehle apni

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database (PDF ya files) se sahi info dhoond kar dete hain (Retrieve), aur phir kehte hain ke ab is info ko use kar ke jawab generate karo. Is se AI "hallucinate" nahi karta.

8. LangChain

Ye ek Frame hai jo in sab cheezon ko aapas mein jorti hai. Agar aap ne ek app banani hai jis mein LLM bhi ho, VectorDB bhi ho aur RAG bhi, toh LangChain wo glue hai jo in sab ko connect kar ke ek poora system banati hai.